



East West University

Department of Computer Science and Engineering

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Course Title : Electrical Circuits

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Section: 03

Experiment No : 04

Experiment Title: Bias Point Detail Analysis of DC Circuit With Dependent Sources Using PSpice Schematics

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1. To analyze Bias Point Detail of DC circuit with dependent source using PSpice Schematics.

[illegible]

1. Here ,

$$I_S = i_1$$

$$V_{ab}=8i_2$$

Applying KVL at super mesh,

$$8i_1 + 20i_2 + 8i_2 = 10 - 0.5I_s$$

$$8.5i_1 + 28i_2 = 10 \dots\dots (1)$$

From supermesh,

$$i_2 - i_1 = 0.5V_{ab}$$

$$i_2 - i_1 = 0.5 \times 8i_2$$

$$i_2 - i_1 - 4i_2 = 0$$

$$i_1 - 3i_2 = 0 \dots\dots\dots (2)$$

From Equation 1 & 2 we get,

$$i_1 = -12 \text{ A}$$

$$i_2 = 4 \text{ A}$$

$$I_s = i_1 = -12 \text{ A}$$

$$V_{ab} = 8i_2 = 8 \times 4 = 32 \text{ V}$$

2. As the experiment was done in the PSpice, so there is no discrepancy in the experimental values.

The experimental values of the voltages and the currents are exactly same with the theoretical values.

Conclusion:

In this experiment, we learnt about 4 types of dependent sources which are VCVS, CCCS, VCCS & CCVS. We used bubble for easier interconnections. And learnt how to do mesh analysis. Also there was no discrepancy as the experiment was done in PSpice.

